



Traveling wave solutions of Burgers' equation with time delay



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ABSTRACT

We study the occurrence of traveling wave solutions for a modified Burgers' equation with diffusivity D and a time delay τ . Monotonic traveling waves with speed c are shown to exist when $\tau c^2/D \leq e^{-1}$. Graphs are provided to help characterize the behavior of these solutions compared to those for the familiar Burgers' equation without delay.

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1. Introduction

We begin by defining Burgers' equation with a time delay. It is a parabolic partial differential equation (PDE),

$$\frac{\partial u}{\partial t}(x, t + \tau) + u(x, t) \frac{\partial u}{\partial x}(x, t) = D \frac{\partial^2 u}{\partial x^2}(x, t), -\infty < x < \infty, t \geq 0, \\ D > 0, \tau > 0 \text{ constants.} \quad (1)$$

This PDE, in particular with the delay in the first term, may be used to model the discrete underlying atomic structure of a fluid medium through which a wave is propagating.

Other variants of this equation have also been treated ([1,2]). The current version has been studied before and an exact solution, periodic in the delay, was presented in [3]. The exact solution was related to that of the heat equation. (Also an instability was uncovered in the analysis of the heat equation with time delay when solved on a finite spatial interval [4]). The exact solution in [3] was not of the form desired here in that it was not a *traveling wave* [5] solution which we will shortly define.

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Owing to the delay, we pose an atypical initial-boundary value problem (IBVP) for (1). Let the initial function be

$$u(x, t) = u_0(x, t), -\infty < x < \infty, 0 \leq t \leq \tau, \quad (2)$$

where $u_0(x, t)$ is a bounded, smooth function and tends to 0 as $x \rightarrow \infty$ and to $2c$ as $x \rightarrow -\infty$. The initial domain is a doubly infinite strip in the $x-t$ plane. The initial function $u_0(x, t)$ leads to a traveling wave.

Look for *traveling wave* solutions using the traveling wave substitution $u = f(x - ct)$.

We have

$$u(x, t + \tau) = f(x - c(t + \tau)) = f(x - ct - c\tau)$$

Hence

$$\frac{\partial u}{\partial t}(x, t + \tau) = -cf'(x - ct - c\tau).$$

Substituting into the PDE (1) find

$$-cf'(x - ct - c\tau) + f(x - ct)f'(x - ct) = Df''(x - ct). \quad (3)$$

Let $\xi = x - ct$. Then (3) becomes

$$-cf'(\xi - c\tau) + f(\xi)f'(\xi) = Df''(\xi). \quad (4)$$

Normalize the amplitude by the wave speed:

$$f(\xi) = 2cg(\xi),$$

so that

$$\begin{aligned} f'(\xi) &= 2cg'(\xi) \\ f''(\xi) &= 2cg''(\xi). \end{aligned}$$

Thus

$$f'(\xi - c\tau) = 2cg'(\xi - c\tau),$$

and (4) becomes

$$-c(2cg'(\xi - c\tau)) + 2cg(\xi)(2cg'(\xi)) = D(2cg''(\xi)).$$

So for $c \neq 0$

$$-g'(\xi - c\tau) + 2g(\xi)g'(\xi) = \frac{D}{c}g''(\xi). \quad (5)$$

Integrate (5).

We are interested in solutions of

$$k \frac{dg}{d\xi}(\xi) = g^2(\xi) - g(\xi - c\tau), \quad (6)$$

where $c > 0, k = D/c > 0, \tau > 0$ such that

$$g \rightarrow 0, g' \rightarrow 0 \text{ as } \xi \rightarrow \infty \text{ and } g \rightarrow 1, g' \rightarrow 0 \text{ as } \xi \rightarrow -\infty. \quad (7)$$

The resulting equation (6) is a classical *differential-delay equation* (DDE).

In the next section, we derive an ordinary differential equation (ODE) satisfied by solutions to (6)–(7) by means of an integral equation suggested by the IBVP, and later observe the graphical appearance of these solutions.

In the last section, these traveling wave solutions are put in the context of the PDE with delay. A conservation law is derived, similar to the case of no delay.

2. Derivation of the ordinary differential equation

The simplest way to convert (6) to an ordinary differential equation is to integrate it from $-\infty$ to ξ to obtain an integral equation

$$\begin{aligned} kg(\xi) - k &= \int_{-\infty}^{\xi} g^2(s) ds - \int_{-\infty}^{\xi} g(s - c\tau) ds \\ &= \int_{-\infty}^{\xi} g^2(s) ds - \int_{-\infty}^{\xi - c\tau} g(s') ds' \quad (s' = s - c\tau) \\ &= \int_{-\infty}^{\xi} g^2(s) ds - \int_{-\infty}^{\xi} g(s) ds + \int_{\xi - c\tau}^{\xi} g(s) ds. \end{aligned} \quad (8)$$

By (2), the last term on the right of (8) is known. Define

$$\frac{1}{k} \int_{\xi - c\tau}^{\xi} g(s) ds + 1 := \Phi(\xi, \tau).$$

For instance, suppose

$$u(x, t) = u_0(x, t) = \frac{1}{1 + e^{2(x - ct)/k}} = \frac{1 - \tanh((x - ct)/k)}{2}, \quad -\infty < x < \infty, \quad 0 \leq t \leq \tau. \quad (9)$$

This function in the shape of an inverted sigmoid (shown in Section 2 to follow) is the traveling wave solution to (6) when $\tau = 0, \xi = x - ct$, whose stability was studied in [6]. It satisfies (7). We take this as given so that (8) reads

$$g(\xi) = \frac{1}{k} \int_{-\infty}^{\xi} [g^2(s) - g(s)] ds + \frac{1}{k} \int_{\xi - c\tau}^{\xi} \frac{1}{1 + e^{2s/k}} ds + 1, \quad (10)$$

which is a nonlinear Volterra integral equation whose solution is known to exist [7]. Its existence leads to the solution of (6) and hence of (1).

One way to handle this is to simply differentiate both sides of (10) with respect to ξ , obtaining by Leibniz' rule and (9):

$$k \frac{dg}{d\xi} = g^2(\xi) - g(\xi) + \frac{1}{2} \tanh((\xi - c\tau)/k) - \frac{1}{2} \tanh(\xi/k). \quad (11)$$

So we have converted the DDE into an ODE for $g(\xi)$ with delayed forcing τ . Also needed is a suitable initial condition say $g(0)$.

We now examine in more detail the linearized behavior of the fixed point $g_0 = 0$ of (6) as $\xi \rightarrow \infty$. The explicit representation of the fundamental solution $K(\xi)$ of the linear DDE

$$k \frac{dg}{d\xi}(\xi) = -g(\xi - c\tau) \quad (12)$$

is known (see reference [8]). Its characteristic equation found by setting

$$g = e^{\lambda \xi}$$

is given by

$$k\lambda = -e^{-\lambda c\tau}. \quad (13)$$

For

$$c\tau k^{-1} \leq e^{-1},$$

there are real eigenvalues λ^* , which are given by

$$\lambda^* = \frac{W(-c\tau/k)}{c\tau},$$

where $W(x)$ is Lambert's W -function [9]. See Fig. 1.

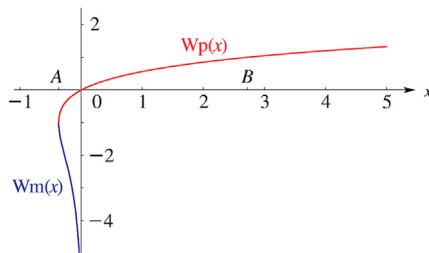


Fig. 1. Branches $W_p(x)$ and $W_m(x)$ of the Lambert W -function. A and B denote the points $-1/e$ and e , respectively, on the x -axis. On the x -interval $(-1/e, 0)$, there are two real solutions. [10, Figure 4.13.1].

There are also complex eigenvalues. The characteristic equation (13) has a solution $\lambda = \alpha + i\beta$ if

$$e^{-c\tau\alpha} \cos(\beta c\tau) + k\alpha = 0$$

and

$$e^{-c\tau\alpha} \sin(\beta c\tau) = k\beta$$

It can be proved [11, Chapter 13] that all of the eigenvalues have negative real parts **if and only if**

$$\frac{c\tau}{k} < \frac{\pi}{2}.$$

However, within this range, eigenvalues are only complex for $e^{-1} < c\tau/k < \pi/2$, which is related to oscillatory behavior [12]. Thus initially small perturbations decay, which connotes asymptotic stability. Monotonic decay will occur for $c\tau k^{-1} \leq e^{-1}$. This will reflect the Lyapunov stability of such a fixed point [11, Theorem 11.4].

We are also interested in the behavior as $\xi \rightarrow -\infty$.

In this case set

$$h(\xi) = 1 - g(\xi).$$

Then (6) becomes

$$kh'(\xi) = -h^2(\xi) + 2h(\xi) - h(\xi - c\tau).$$

Using the substitution

$$h = e^{\lambda\xi},$$

in the linearized equation near the fixed point $h_0 = 0$, find

$$k\lambda = 2 - e^{-\lambda c\tau}.$$

By a similar analysis as before, monotonic increase as $\xi \rightarrow -\infty$ will occur over the parameter range $c\tau k^{-1} \leq e^{(-1+2c\tau k^{-1})}$. The same result on Lyapunov stability [11, Theorem 11.4] will apply, this time over the interval $-\infty < \xi \leq 0$. Since $e^{-1} \leq e^{(-1+2c\tau k^{-1})}$, the traveling wave behavior on $-\infty < \xi < \infty$ follows thereby for $c\tau k^{-1} \leq e^{-1}$.

2.0.1. Global monotonicity

The notion of a monotonic solution of (8) on $(-\infty, \infty)$, in this case, positive, decreasing and bounded below by zero, will depend on whether successive approximations $g_n(\xi)$ are *monotonic decreasing* in this way. For instance by a standard iteration argument,

$$g_{n+1}(\xi) = \Phi(\xi, \tau) + \frac{1}{k} \int_{-\infty}^{\xi} [g_n^2(s) - g_n(s)] ds, n = 0, 1, 2, \dots$$

when a unique solution exists [11, Chapter 7]. Then if $\Phi(\xi, \tau)$ is monotone decreasing in ξ , $0 < \Phi(-\infty, \tau) < \infty$, $g(\xi)$ is monotonic decreasing whenever $[g^2(\xi) - g(\xi)]$ has this property. This is true for (6) and (11) when $c\tau k^{-1} \leq e^{-1}$.

2.1. Graphical representations of the solutions

Figs. 2 and 3 show realizations of the solutions to (11) by use of Maple 2018 with the command *dsolve*. The initial condition is $g(0) = 0.5$.

We have taken $c = k = 1$. The reader will observe that the graphs of g for $\tau = 0$ to $\tau = 0.3678 \simeq e^{-1}$ are monotonic, while when $\tau = 0.5$, the graph is not monotonic.

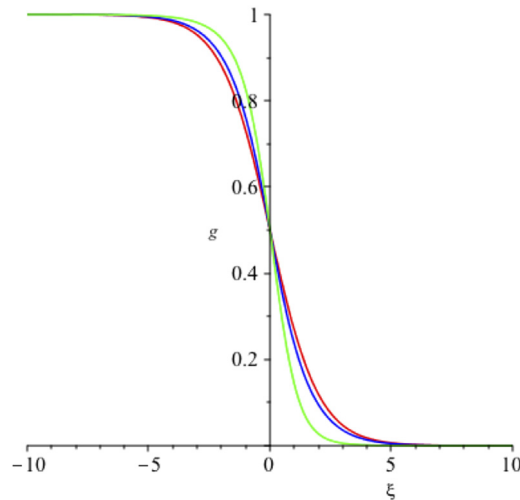


Fig. 2. Red $\tau = 0$, blue $\tau = 0.1$, green $\tau = 0.3678$.

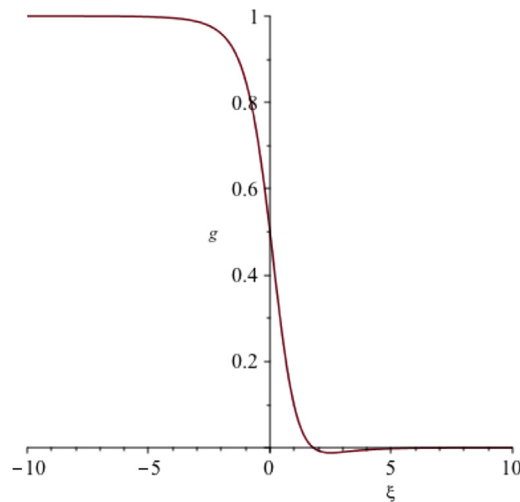


Fig. 3. $\tau = 0.5$.

3. Asymptotic persistence of traveling waves

Using the approach of Peletier [6] for the classical Burgers equation we asserted the following.

Define an initial-boundary value problem (IBVP) for (1). Let the initial function be

$$u(x, t) = u_0(x, t), -\infty < x < \infty, 0 \leq t \leq \tau, \quad (14)$$

where $u_0(x, t)$ is a bounded, smooth function and tends to 0 as $x \rightarrow \infty$ and to $2c$ as $x \rightarrow -\infty$. The parametric condition assumed for this to hold is $c\tau k^{-1} \leq e^{-1}$.

I. Then there exists a smooth solution to (1), (2), given by $u(x, t) = 2cg(x - ct)$, from (6) and (11).

II. Let u and v be two distinct smooth solutions of the PDE (1), where $u(x, t)$ satisfies (14), while $v(x, t)$ satisfies

$$v(x, t) = v_0(x, t), -\infty < x < \infty, 0 \leq t \leq \tau. \quad (15)$$

Then

$$\int_{-\infty}^{\infty} (u - v) dx = \text{const.} \quad (16)$$

To see this consider the equation for $z(x, t) = u(x, t) - v(x, t)$, found from (1):

$$\frac{\partial z}{\partial t}(x, t + \tau) + \frac{1}{2} \frac{\partial}{\partial x}(u^2 - v^2) = D \frac{\partial^2 z}{\partial x^2}. \quad (17)$$

Integrate (17) over the x -interval $(-\infty, \infty)$. We find

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} z(x, t + \tau) dx + \frac{1}{2} [z(x, t)(u(x, t) + v(x, t))]_{-\infty}^{\infty} = \left[D \frac{\partial z(x, t)}{\partial x} \right]_{-\infty}^{\infty}.$$

Thus

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} (u(x, t + \tau) - v(x, t + \tau)) dx = 0.$$

Then (16) follows for all t .

This generic conservation law matches, to a great extent, the behavior of the stable traveling waves of Burgers' equation without delay [6].

4. Conclusions

We have provided a traveling wave solutions analysis of Burgers' equation with a time delay. The approach reduces the problem to an ordinary differential equation with a forcing term. Special solutions were found that are monotonic decreasing in the wave argument $\xi = x - ct$. Figures were provided, describing these traveling waves for various values of the delay. By closely observing the equation without delay, we were able to conclude that most of the same properties, of stability and limiting behavior as $\xi \rightarrow \pm\infty$, are retained when the time delay τ , diffusivity D and wave speed c satisfy the condition $\tau c^2/D \leq e^{-1}$.

CRedit authorship contribution statement

Isom Herron: Formal analysis, Writing. **Clement McCalla:** Investigation. **Ronald Mickens:** Conceptualization.

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